

WILMINGTON NWS, OH, USA

WMO: 724296

Lat: 39.4214N

Lon: 83.8217W

Elev: 325

StdP: 97.48

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 13841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.8	-20.1	0.7	-14.9	-17.3	0.9	-12.1	12.4	2.3	11.4	2.0	3.1	270	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.2	23.2	31.0	22.8	29.6	22.1	24.9	30.0	24.1	28.7	23.4	27.6	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	18.8	27.4	22.6	18.0	26.6	22.1	17.5	26.0	77.3	30.1	74.1	28.9	71.1	27.5	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	8.2	-20.1	33.8	3.8	1.7	-22.9	34.9	-25.1	35.9	-27.2	36.9	-29.9	38.1		
(5)			-20.2	26.2	3.9	0.9	-22.9	26.9	-25.2	27.5	-27.4	28.0	-30.2	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-1.7	0.0	5.2	11.8	17.4	21.7	23.1	22.4	19.1	12.6	6.3	1.1		
	DBStd	10.12	6.89	6.39	6.15	5.02	4.42	3.24	2.81	2.69	3.90	5.00	5.33	5.92		
	HDD10.0	1311	366	283	176	38	3	0	0	0	0	29	135	280		
	HDD18.3	2946	621	512	409	200	72	9	1	3	37	188	360	533		
	CDD10.0	1915	3	4	26	93	233	351	407	383	273	110	25	6		
	CDD18.3	509	0	0	1	5	44	110	150	128	60	11	0	0		
	CDH23.3	4194	0	0	8	47	352	888	1246	1016	548	89	1	0		
	CDH26.7	1278	0	0	1	3	72	276	426	313	171	17	0	0		
(14) Wind	WSAvg	3.7	4.7	4.5	4.4	4.5	3.6	3.3	2.8	2.6	2.8	3.5	4.0	4.3		
(15) Precipitation	PrecAvg	1079	83	65	94	97	137	110	96	85	77	80	70	85		
	PrecMax	1466	186	134	217	205	282	196	170	146	163	154	148	208		
	PrecMin	658	35	25	46	29	40	47	38	37	28	26	18	27		
	PrecStd	163	35	33	39	45	67	44	30	32	45	42	33	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.9	24.0	27.1	30.4	32.7	34.0	33.4	32.8	28.8	22.2	17.6		
		MCWB	14.5	15.3	16.1	18.3	21.4	22.6	24.5	23.2	21.6	20.1	15.7	15.0		
	2%	DB	14.0	15.1	20.3	24.8	28.4	31.1	32.2	31.4	30.7	25.8	19.1	14.7		
		MCWB	12.2	12.5	14.1	16.4	20.3	22.4	24.1	23.1	21.6	18.6	14.1	12.6		
	5%	DB	11.8	12.6	17.4	22.6	27.3	29.6	30.8	29.9	28.6	23.3	17.2	12.4		
		MCWB	10.2	10.0	12.4	15.5	19.8	21.7	23.3	22.5	20.9	17.0	13.5	10.6		
	10%	DB	8.4	9.6	15.0	20.5	25.4	28.1	29.1	28.3	26.7	21.1	14.8	9.9		
		MCWB	6.5	6.9	11.3	14.4	18.8	21.1	22.7	21.8	19.5	16.5	11.4	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.6	17.3	20.0	23.0	25.0	26.4	25.5	24.1	21.8	17.4	15.7		
		MCDB	16.7	18.5	23.1	25.5	28.0	30.3	31.3	30.4	29.7	27.3	19.8	17.3		
	2%	WB	12.7	12.9	15.3	18.0	21.9	23.9	25.2	24.7	23.0	20.2	15.6	13.1		
		MCDB	14.0	14.4	18.9	22.2	26.6	28.5	30.5	29.4	27.8	23.7	18.2	14.1		
	5%	WB	10.3	10.3	13.7	16.6	20.8	23.0	24.3	23.8	22.0	18.6	13.7	10.8		
		MCDB	11.5	12.4	16.6	20.7	25.2	27.7	29.0	27.9	26.4	22.2	16.4	12.0		
	10%	WB	6.5	6.9	11.6	15.4	19.8	22.2	23.5	22.9	21.0	16.9	11.9	8.0		
		MCDB	8.2	9.3	14.7	19.2	24.0	26.6	27.7	26.4	24.5	20.3	14.5	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.8	9.9	11.4	10.7	10.6	10.6	11.1	12.0	10.9	9.6	7.5		
		MCDBR	10.3	12.4	12.9	13.4	12.2	12.3	11.9	12.5	14.1	13.1	12.1	10.1		
	5% WB	MCWBR	9.1	10.1	8.2	7.9	6.0	5.4	5.0	5.7	6.2	7.0	8.4	8.8		
		MCWBR	9.6	11.6	11.2	11.5	10.6	10.3	10.5	10.3	11.4	10.9	10.5	9.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.301	0.321	0.347	0.388	0.434	0.445	0.451	0.449	0.392	0.352	0.322	0.301		
		taud	2.463	2.370	2.375	2.273	2.177	2.195	2.182	2.170	2.311	2.423	2.469	2.470		
	Ebn at Noon	Ebn at Noon	873	898	909	888	849	837	828	819	849	851	838	843		
		Edh at Noon	79	100	110	130	147	144	145	143	117	94	79	73		
(44) All-Sky Solar Radiation	RadAvg		1.75	2.57	3.58	4.84	5.51	6.07	5.87	5.46	4.48	3.19	2.17	1.48		
	RadStd		0.16	0.32	0.31	0.39	0.46	0.37	0.34	0.30	0.43	0.30	0.26	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+72
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+70

Nomenclature: See separate page